Technical Assessment Report: Cynbit Project Planner

Full Stack Developer Intern - Rashi Verma

Cynbit Technologies Private Limited (cynbit.in)

Repository: https://github.com/RashiVerma14/cynbit-project-planner

Live Demo: https://rashiverma14.github.io/cynbit-project-planner/

Role: Full Stack Developer Intern
Applicant: Rashi Verma
Project: Cynbit Project Planner — *"From Business Vision to Technical Blueprint"*
Target Organization: Cynbit Technologies Private Limited ([cynbit.in](https://www.cynbit.in/))
Date: November 2026

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1. Executive Summary & Website Analysis

Cynbit Technologies is an IT services and digital transformation company headquartered in Jaipur, Rajasthan, specializing in autonomous AI agents, enterprise SaaS systems (*TenantEase*, *Revio*, *Cynbit Studio*), and technical education (*BitElevate*).

### Analysis of the Existing Website (User & Developer Perspective)
* **Visual & Aesthetic Strengths:**
  * Clean, futuristic dark-mode aesthetic utilizing deep obsidian slate (`#07090E`), electric cyan (`#06B6D4`), and deep indigo (`#6366F1`) gradients.
  * Modern card typography and fine sub-pixel borders (`border-white/10`) convey engineering rigor and technical precision.
* **The Strategic Problem:**
  * **Asymmetric Technical Information:** Prospective clients visiting Cynbit with innovative digital or AI ideas frequently lack clarity regarding feature feasibility, appropriate technology stacks, and development effort.
  * **High Initial Friction:** Visitors are immediately channeled toward a standard "Contact Us" or consultation inquiry form. Without an interactive self-service mechanism, many potential leads drop off before initiating contact.
* **The Proposed Solution:**
  * Build the **Cynbit Project Planner**, an interactive discovery platform that allows prospective clients to define their requirements in a structured 7-step wizard and instantly receive an indicative engineering estimate, a tailored technology stack, and an AI-synthesized architectural blueprint.

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2. System Architecture & Component Design

The platform follows a modern decoupled full-stack architecture:

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```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                Client Layer                 %
%         React 19 + Vite + Tailwind CSS          %
%  (Landing Page, 7-Step Wizard, Results HUD, Admin UI)  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%,%%%%%%%%%%%%%%%%%%%%%%%%%%
                   %  HTTP / JSON (Axios)
                   %¼
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                API Gateway                  %
%         Express.js + Security Middleware          %
%    (CORS, Centralized Error Handling, JWT Auth)      %
%%%%%%%%%%%%%%,%%%%%%%%%%%%%,%%%%%%%%%%%%%,%%%%%%%%%%%%%
          %          %          %
          %¼         %¼         %¼
%%%%%%%%%%%%%%%%%%%  %%%%%%%%%%%%%%%  %%%%%%%%%%%%%%%%%%%%
%   Estimation   % %  Tech Stack  % %   Groq AI Engine   %
%     Service    % %  Recommender % % (Llama 3.3 70B JSON)%
~~~~

```
%    (Math Engine)   %  %     Engine      %  %    + Resilient Rules %
%%%%%%%%%%%%%,%%%%%  %%%%%%%%,%%%%%%%  %%%%%%%%%%%%%%%,%%%%%%%%%
            %              %                    %
            %%%%%%%%%%%%%%<%%%%%%%%%%%%%%%%%%%%
                           %¼
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                    Persistence Layer                   %
%              MongoDB Atlas (Mongoose ODM)              %
%         (Project Inquiries & Admin Credentials)        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

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3. Technology Stack & Justification

Technology	Selection	Justification
Frontend Framework	React.js (Vite)	High-velocity SPA rendering, declarative component state management, and optimized asset bundling (< 250KB total bundle).
Styling & Motion	Tailwind CSS + Framer Motion	Rapid design token execution matching Cynbit's visual identity with fluid step transitions and accessible micro-interactions.
Icons & Form Logic	Lucide React + React Hook Form	High-performance tree-shakeable icons and performant controlled form inputs.
Backend Runtime	Node.js & Express.js	Asynchronous event loop, low memory footprint, and modular separation of routes, controllers, and services.
Database	MongoDB Atlas & Mongoose	Flexible document model accommodating dynamic capability arrays, polymorphic project attributes, and nested JSON blueprints.
Authentication	JSON Web Tokens (JWT) + bcryptjs	Stateless, cryptographic session validation for administrative lead endpoints with salted password hashing.
AI Synthesis	Groq Cloud API	Ultra-low inference latency (< 1.5s) using LPU acceleration on open-weights foundation models (`llama-3.3-70b-versatile`).

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4. Database Schema Design (MongoDB / Mongoose)

### `ProjectInquiry` Collection
Captures the entire project specification and generated blueprint:
* **Identification:** `projectId` (Indexed, unique, format: `CYN-YYYY-XXXX`).
* **Lead Dossier:** `clientInfo` (`fullName`, `businessEmail`, `companyName`, `phone`, `additionalNotes`).
* **Project Scope:** `category`, `projectName`, `description`, `problemStatement`, `targetAudience`, `industry`.
* **Technical Parameters:** `selectedCapabilities` (Array), `complexity` (`BASIC_MVP`, `INTERMEDIATE`, `ADVANCED`, `ENTERPRISE`), `expectedUsers`, `preferredTechnology`.
* **Estimation Output:** `totalEffortHours`, `indicativeCostMinINR`, `indicativeCostMaxINR`, `estimatedDurationWeeks`, `foundationHours`, `frontendHours`, `backendHours`, `testingQaHours`.
* **Architectural Blueprint:** `executiveSummary`, `suggestedArchitecture`, `coreModules`, `phases` array, `scalabilityConsiderations`, `riskMitigation`, `isAiGenerated`.
* **Lead Lifecycle:** `inquiryStatus` (`NEW`, `UNDER_REVIEW`, `CONTACTED`, `IN_DISCUSSION`, `CONVERTED`, `REJECTED`).

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5. Deterministic Estimation Methodology

To eliminate arbitrary or fabricated estimates, the calculation engine applies a transparent Work Breakdown Structure (WBS):

### Mathematical Formulation
$$\text{Total Effort Hours } (H) = \left(E_{\text{base}} + \sum_{i=1}^{n} E_{\text{cap}, i} \right) \times M_{\text{complexity}}$$

1. **Base Category Effort ($E_{\text{base}}$):**
   * AI-Powered Application: $130\text{h}$

  * SaaS Platform: $110\text{h}$
  * Custom Enterprise Software: $150\text{h}$
  * Mobile Application: $85\text{h}$
  * Business Automation: $75\text{h}$
  * Web Application: $70\text{h}$
2. **Capability Adders ($E_{\text{cap}}$):**
  * Autonomous AI Agents: $+45\text{h}$
  * AI Conversational Bot: $+32\text{h}$
  * Admin / Operator Console: $+28\text{h}$
  * Auth & Multi-Tenant RBAC: $+24\text{h}$
  * Payment Gateway & Billing: $+22\text{h}$
  * Real-time WebSockets / Events: $+18\text{h}$
3. **Complexity Multiplier ($M_{\text{complexity}}$):**
  * Basic MVP: $1.0\times$
  * Intermediate Production: $1.35\times$
  * Advanced High-Concurrency: $1.75\times$
  * Mission-Critical Enterprise: $2.25\times$
4. **Financial Banding:**
    $$\text{Budget Range (INR)} = [H \times ¹900, \; H \times ¹1,350]$$
5. **Sprint Duration:**
  $$\text{Duration (Weeks)} = \max\left(3, \left\lceil \frac{H}{36\text{h}} \right\rceil\right)$$

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6. AI Blueprint Integration & Resilience

* **Groq Cloud Integration:** Connects securely on the backend via server-side HTTPS requests, passing sanitized project scope parameters.
* **Structured JSON Enforcement:** Prompts require RFC-compliant JSON matching schema fields (`executiveSummary`, `coreModules`, `phases`, `scalabilityConsiderations`).
* **Deterministic Fallback Engine:** If the Groq API key is absent, network timeouts occur, or rate limits trigger, the system immediately switches to a rule-based architectural synthesizer. This guarantees **100% uptime** and ensures evaluation is never blocked.

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7. Security & Validation Practices

1. **Zero Secret Leakage:** No API keys (Groq, MongoDB, JWT secret) are exposed to the client bundle. All credentials load strictly via server environment variables.
2. **Input Validation:** Middleware sanitizes email addresses, enforces mandatory field checks, and rejects malformed payloads prior to controller execution.
3. **Cryptographic Password Hashing:** Admin passwords use `bcryptjs` with salt rounds of 10.
4. **JWT Bearer Protection:** Admin endpoints require a valid `Bearer <token>` in authorization headers.

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8. Verification & Test Results

The backend includes native Node.js tests verifying core logic:
* `' Estimation Service: Calculates deterministic effort for MVP`
* `' Estimation Service: Scales with complexity tiers`
* `' Recommendation Service: Generates appropriate stack for category`
* `' AI Service: Fallback blueprint provides complete specification`

Test Outcome: 4 passed, 0 failed (150ms execution time).
Frontend Build: Vite production build executed in 1.29s producing optimized, gzipped assets.

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9. Conclusion

The **Cynbit Project Planner** demonstrates that a Full Stack Developer Intern can analyze business friction, design a brand-aligned user interface, implement a resilient backend with algorithmic business logic, and deliver a production-ready application.